



NeuroSpace

Complete evaluation export

Job ID	b9032523-6ea2-426b-9990-157f78712d5a
Status	completed
Result timestamp	2026-06-01T19:13:19.627385
PDF generated	2026-06-01 20:19:09

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0010
mitigation_enabled	True
mitigation_runs	3

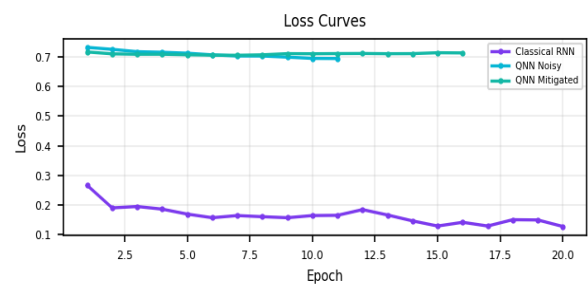
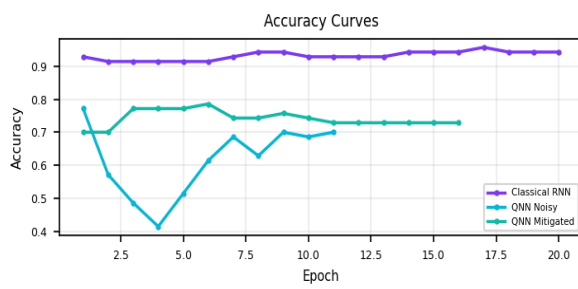
Best Model

Winner	RNN Accuracy	QNN Accuracy	Improvement
RNN	95.00%	81.43%	13.57%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss	Training Time
Classical RNN	95.00%	95.03%	95.00%	95.01%	0.1270	1.89s
QNN Noisy	69.29%	67.64%	69.29%	67.60%	0.6950	40.63s
QNN Mitigated	81.43%	82.29%	81.43%	80.11%	0.7139	1m 58.66s

Learning Curves



Classical RNN

Metric	Value
Accuracy	95.00%
Precision	95.03%
Recall	95.00%
F1 Score	95.01%
Loss	0.1270
Training Time	1.89s
Train Loss	0.0559
Val Loss	0.1270
Train Accuracy	97.96%
Val Accuracy	94.29%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	3	45

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	96.70%	95.65%	96.17%	92.0000
1	91.84%	93.75%	92.78%	48.0000
macro avg	94.27%	94.70%	94.48%	140.0000
weighted avg	95.03%	95.00%	95.01%	140.0000
Accuracy			95.00%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.2362	0.2663	94.27%	92.86%
2	0.1086	0.1898	96.52%	91.43%
3	0.0996	0.1945	97.55%	91.43%
4	0.0918	0.1856	97.34%	91.43%
5	0.0741	0.1688	97.96%	91.43%
6	0.0893	0.1565	96.93%	91.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.0621	0.1639	98.36%	92.86%
8	0.0591	0.1600	98.16%	94.29%
9	0.0575	0.1566	97.96%	94.29%
10	0.0710	0.1639	97.96%	92.86%
11	0.0588	0.1647	97.75%	92.86%
12	0.0747	0.1840	97.55%	92.86%
13	0.0827	0.1658	97.14%	92.86%
14	0.0628	0.1458	98.16%	94.29%
15	0.0500	0.1285	98.57%	94.29%
16	0.0495	0.1411	98.77%	94.29%
17	0.0478	0.1285	98.16%	95.71%
18	0.0700	0.1498	97.55%	94.29%
19	0.0681	0.1491	97.75%	94.29%
20	0.0559	0.1270	97.96%	94.29%

QNN Noisy

Metric	Value
Accuracy	69.29%
Precision	67.64%
Recall	69.29%
F1 Score	67.60%
Loss	0.6950
Training Time	40.63s
Train Loss	0.6730
Val Loss	0.6950
Train Accuracy	76.48%
Val Accuracy	70.00%
Train Size	489
Val Size	70
Test Size	140

QNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	78	14
Class 1	29	19

QNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
0	72.90%	84.78%	78.39%	92.0000
1	57.58%	39.58%	46.91%	48.0000
macro avg	65.24%	62.18%	62.65%	140.0000
weighted avg	67.64%	69.29%	67.60%	140.0000
Accuracy			69.29%	

QNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7185	0.7327	75.66%	77.14%
2	0.7044	0.7259	65.24%	57.14%
3	0.6972	0.7180	48.26%	48.57%
4	0.6910	0.7161	41.31%	41.43%
5	0.6863	0.7130	48.47%	51.43%
6	0.6831	0.7073	73.82%	61.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.6796	0.7035	76.89%	68.57%
8	0.6779	0.7034	74.44%	62.86%
9	0.6770	0.6996	74.03%	70.00%
10	0.6748	0.6953	76.48%	68.57%
11	0.6730	0.6950	76.48%	70.00%

QNN Mitigated

Metric	Value
Accuracy	81.43%
Precision	82.29%
Recall	81.43%
F1 Score	80.11%
Loss	0.7139
Training Time	1m 58.66s
Train Loss	0.6124
Val Loss	0.7139
Train Accuracy	83.03%
Val Accuracy	72.86%
Train Size	489
Val Size	70
Test Size	140

QNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	22	26

QNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
0	80.00%	95.65%	87.13%	92.0000
1	86.67%	54.17%	66.67%	48.0000
macro avg	83.33%	74.91%	76.90%	140.0000
weighted avg	82.29%	81.43%	80.11%	140.0000
Accuracy			81.43%	

QNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6850	0.7172	71.78%	70.00%
2	0.6644	0.7108	79.96%	70.00%
3	0.6510	0.7093	83.23%	77.14%
4	0.6428	0.7094	83.03%	77.14%
5	0.6358	0.7076	84.46%	77.14%
6	0.6309	0.7063	84.05%	78.57%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.6265	0.7059	84.25%	74.29%
8	0.6245	0.7075	84.05%	74.29%
9	0.6220	0.7115	83.44%	75.71%
10	0.6206	0.7111	83.84%	74.29%
11	0.6188	0.7116	84.05%	72.86%
12	0.6181	0.7120	83.64%	72.86%
13	0.6170	0.7112	83.44%	72.86%
14	0.6180	0.7115	82.82%	72.86%
15	0.6172	0.7145	83.23%	72.86%
16	0.6124	0.7139	83.03%	72.86%